



Web-based remote voice control of robotized cells

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ABSTRACT

The main goal of research presented in this paper was to develop an effective tool for remote voice control of robotized cells. This goal has been achieved by proper combination of two techniques: automatic speech recognition and web-based remote control of robotized cells. Comprehensive analysis of existing solutions related to the control with voice commands and those related to the web-based remote control has been performed. Usually the above mentioned topics are dealt with separately. Only few works tried to cope with the problem of how to combine solutions in these two areas. None of them was aimed at industrial applications. Unlike those works, my research is focused on remote voice control of industrial robotized cells. Besides, my approach does not presume application of a small set of simple pre-defined voice commands but it is based on the use of quasi-natural language instead. The main result of this research was the architecture of industrially oriented remote voice control system. Theoretical considerations have been practically verified in the laboratory. According to my knowledge, this laboratory implementation is the first-ever industrially oriented operational remote voice control system based on commands in quasi-natural language.

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1. Introduction

The current paper presents research conducted by the author in order to develop an effective tool for remote, web-based voice control of robotized cells. To achieve this goal, it was necessary to combine two techniques: automatic speech recognition and web-based remote control of robotized cells. Therefore a comprehensive analysis of problems and existing solutions related to both areas has been performed.

Each of those research domains has its own challenges, solved and unsolved problems, therefore they are usually dealt separately. Human-machine voice communication has been the subject of research for many years. The most popular objects controlled by voice are mobile robots, particularly assistive robots [1–12]. There are only a handful of works dealing with industrial applications of automatic speech recognition [13–15]. Based on the literature survey contained in Section 2, I will try to answer the question: when is the use of automatic speech recognition advisable in industrial robotized systems? What conditions must be fulfilled in order to provide an effective and errorless voice communication? The answers to these questions will be later the basis for similar questions and answers regarding remote voice control.

The use of Internet in industry covers a very broad spectrum of issues: from various aspects of design [16] through product and production data exchange [17,18], process planning [19] to the remote control of industrial equipment [20–25]. Many web-based applications are related to the rapid prototyping and manufacturing (RP&M) technique [26–29].

As far as the web-based remote control is concerned, the majority of works deal with industrial or mobile robots [20–22,30–35], although applications oriented on CNC machine tools [23,24] or even on the whole machining cells [25] have been reported, too. Based on the literature survey contained in Section 3, I will try to answer the question: what are the limitations and requirements resulting from the fact that the robotized cell is controlled remotely? What conditions must be fulfilled in order to provide errorless and effective remote control? The answers to these questions will also determine conditions that should be imposed on the voice command processing scheme in remote voice control applications.

Combining the conclusions resulting from the analysis of problems and solutions related to both domains, I will try to determine the optimal remote voice control scenario as well as appropriate remote voice control system architecture. Such a comprehensive approach seems to be novel. Although many scientific papers present research in the area of distributed speech recognition [36–39], they usually do not deal with voice control. Only very few works present solutions related to combination of voice control and remote control. None of those solutions are industrially oriented ones. None of them are based on voice commands in quasi-natural language.

The remainder of this article is organized as follows. Section 2 discusses the problems, challenges and solutions related to the

Abbreviations: CFG—, context-free grammar; NLP—, natural language processing; RP&M—, rapid prototyping and manufacturing; SAPI—, Microsoft speech application programming interface; SR—, speech recognition; VCD—, voice command language description format

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industrially oriented voice control systems (apart from the remote control aspect). Section 3 contains a review of research related to the web-based remote control. Preliminary conclusions regarding combination of voice control and remote control techniques are drawn in Section 4. Architecture of industrially oriented voice control system that can be combined with web-based remote control is proposed in Section 5. Functioning of web-based distributed voice control is discussed in detail in Section 6. Section 7 presents laboratory implementation: the web-based voice control system of an educational robotized cell. Experimental results are discussed in Section 8. Finally, Section 9 contains conclusions and plans for the future.

2. Industrially oriented voice control systems

As stated before, industrial applications of speech recognition are still a minority. Many reasons may contribute to this fact. Probably the most important two of them are as follows:

- Lack of interest, resulting from conviction that voice communication is not a suitable way for the human–machine interaction in industrial environment.
- Poor reliability and effectiveness of existing industrially oriented laboratory prototypes of voice control systems.

Approach presented in the first point does not seem to be objectively justified. Many authors have already proven that. For example, practical and scientific arguments for man-machine voice communication have been presented by Kulyukin [7]. As far as potential industrial applications are concerned, Pires [40] underlines the usefulness of voice communication in industrial robotic cells in the future scenario, where humans and robots safely share the workspace. Synergetic effects resulting from direct cooperation of humans and industrial robots are presented by Woern [41], Lepratti [42] and Lenz et al. [43]. Speech communication is the most natural way to provide such cooperation. Therefore it is not the question whether the voice control systems could be applied in the industry. We should rather ask the question about situations when their use is advisable or even desired. Potential areas of application are determined by the following advantages of speech communication:

- It does not engage operator's hands.
- It does not restrict operator's mobility (in contradistinction to keyboards, buttons and other interaction mechanisms that require the operator to be present at a specific place).
- Uttering a voice command is often more simple and quick than entering data using keyboard (particularly, when input data is very complex).

Therefore a very promising application area of voice control are industrial manufacturing cells of diversified automation level, where some functions are fully automated but there are also actions that must be taken manually by the operator [15]. The operator, who is focused on his own task (and having his hands engaged in manual work), could simultaneously give instructions to his “co-workers” i.e. to machine tools, robots, clamps etc. The use of a wireless microphone would provide the operator with almost unrestricted mobility. This is of particular importance when he/she should supervise an industrial cell consisting of several machines and/or devices.

Poor reliability and low effectiveness of voice control system can be serious factors hindering its implementation. If the speech recognition system were not reliable enough, then wrong (or even dangerous) actions could be performed by robots and/or machines. On the other hand, even a very reliable speech recognition system

would not be useful if it were not user-friendly e.g. if the user were required to memorize the set of possible voice commands instead of communicating in a natural (or at least a quasi-natural) language.

How to provide both reliability and user-friendliness? In order to answer this question, it is necessary first to ask the question about specific features of industrially oriented voice control systems. As stated by Rogowski [15], some requirements that must be fulfilled by them are more restrictive than those related to the non-industrial solutions:

- Correct recognition of all the words constituting a voice command is particularly important.
- Processing of commands that contain the verbal expressions representing numerical values must be provided.
- Reaction to the voice command must be usually instant.

On the other hand, some requirements, significant in other domains, are not so restrictive in industrial voice control applications:

- As the users of those applications will be qualified machine operators, they can be expected to adapt themselves to some restrictions regarding voice command structure. They can also be expected to keep some discipline in speaking and avoid at least some of spontaneous speech effects like repetitions and insertions. Therefore a quasi-natural command language can be used instead of the natural language.
- Although the number of possible command formulations may be very high, the number of corresponding qualitatively different actions to be performed by an industrial machine is usually quite low.

These remarks, together with arguments presented by other authors, determine the conclusions regarding industrially oriented voice control system. Solutions using a relatively small set of strictly defined commands [13,14] do not take into consideration the diversity of command formulations characteristic for natural way of speaking (but they are very reliable because pattern matching instead of pattern recognition is used for voice command processing). On the other hand, it has been proven that the techniques commonly used in NLP are not suitable for voice control. Kulyukin claims that speech recognition algorithms are not reliable enough to provide errorless robot control with voice commands in natural language [2,3]. Therefore he proposes sublanguage acquisition as an alternative. Stiefelhausen et al. [4] state that n-gram language models, commonly used in NLP, are not suitable for small domains where only little domain dependent data is available for training. Rogowski [15] shows that ignoring or “guessing” unrecognized words is not acceptable in industrially oriented voice control systems because those words may radically change the meaning of the whole command. Therefore the use of grammar-based command language definition could be a good solution. However, as stated by Lauria [44], the main consideration is generating a grammar with sufficiently wide coverage for the domain in question. Luckily, the above mentioned low number of qualitatively different actions that can be performed by an industrial machine makes it possible to use CFG for definition of a quasi-natural voice command language. VCD format proposed by Rogowski [15] facilitates development of voice control systems that fulfill all the requirements:

- As the language definition is grammar-based (CFG), high recognition accuracy is granted because SR engine can be “informed” (at every stage of command recognition) which words are expected next. Therefore the wrong hypotheses can be rejected by SR engine.
- VCD provides common structures for syntax and semantics description. Because description of semantics is arithmetically

oriented, it is easy to define commands containing expressions representing numerical values.

- Instant reaction to the voice command is possible because, as shown by Rogowski [15] CFG-based syntax description usually lets recognize the end of spoken command instantly (in contradistinction to statistical methods used in NLP) even if there are pauses in operator's utterance. However, in situations when ambiguity occurs, so-called "execution words" proposed by Wasfy et al. [45] can be applied.
- VCD format makes it possible to handle some spontaneous speech effects like hesitations and self-corrections.

Therefore it seems reasonable to use VCD-based language description also in remote voice control systems. The main advantages of VCD in comparison with other formalisms like e.g. XML used in SAPI are following:

- Due to the application of so called data reduction principle [15], the sublanguage semantics is entirely determined by VCD description. There is no need to create and implement a separate semantic analysis algorithm for each individual voice control application. This is of particular importance in the case of industrially oriented voice control systems where separate sublanguages must be usually defined for controlling individual machines/robots/robotized cells.
- VCD provides much more transparent and compact language description. Therefore it is easy to define sublanguages for individual voice control applications.
- VCD architecture makes it possible to use the libraries of phrases that can be created independently by various authors at different time [15].
- Description of semantics is arithmetically oriented. This is of particular advantage in engineering applications.

3. Web-based remote control systems

Based on the literature review, we will try to answer the question: what are the limitations and requirements resulting from the fact that the robotized cell is controlled remotely? What conditions must be fulfilled in order to provide errorless and effective remote control?

Possible failures of Internet connection may cause serious problems for the web-based control of the robotized cells. Restricted bandwidth and unpredictable transmission delays may constitute a big impediment, too [30,33]. Many authors tried to cope with these problems. Wang et al. [46] propose the remote control system architecture which can guarantee the non-distortion transfer of control information as well as minimization of Internet data transfer time delay. You et al. [20] describe an application of so-called "predictive simulation". Its aim is to check the correctness of robot commands before they are sent over Internet to the real robot. They also present a method for minimization of time delays in transmission of feedback video stream. Wang and Liu [32] claim that direct control is not suitable for Internet-based mobile robot teleoperation. They propose to use telecommanding that should be event-based and not time-based. The robot must be equipped with "local intelligence" in order to handle autonomously unexpected events. This idea has been formulated in various forms by many other researchers. Winfield and Holland state that low-level functions should be run entirely within the robot [31]. The need of high degree of robot autonomy is postulated also by Huosheng et al. [33]. Alvares and Ferreira [23], who deal with remote control of CNC turning center, claim that control actions should be executed locally to provide the safety. Saucy and Mondada [47] state that robot

autonomy and reactive behaviors enable it to navigate and cope with uncertainty in the real world.

A separate topic, discussed in many scientific papers, is the visual feedback that should be provided in order to inform the remote user about the current state of the controlled object and its environment [20,23,30,33]. It can be hardly imaginable to run an industrial remote control system without appropriate visual feedback. However, sending camera images of the shop floor over Internet is bandwidth-consuming. As stated by Wang et al. [24], reducing network traffic and increasing system performance are the major concerns in web-based system developments. Therefore they propose an interesting solution consisting in representing the machines and devices by a scene graph-based Java 3D model at a client side. This model can be updated according to the behavior of physical devices and rendered by the local CPU thanks to communication with those devices using low-volume message passing. Nevertheless, it may be sometimes necessary to use the real web camera to capture unpredictable scenes for diagnostic purposes.

Summarizing this literature review, it can be stated that following aspects must be necessarily taken into account in the remote control of the robotized cells:

- Remote control system architecture should provide the minimization of the network traffic.
- The robotized cell must be characterized by high degree of autonomy. It must be equipped with some "local intelligence".
- Telecommanding should be event-based and not time-based. Therefore it is preferable that the remote user is offered task-oriented options corresponding to complex actions taken by machines.
- Some form of visual feedback must be provided.

These conclusions will also refer to the web-based voice control systems.

Combination of remote control and voice control was the subject of few works only. The work of Wang and Liu [32] can be treated as a step towards it. Admittedly, they do not deal with speech-based remote control, but their approach to telecommanding, based on linguistic commands, could be extended in order to include automatic speech recognition.

Wasfy et al. [45] describe virtual environment for training users in the operation of complex engineering systems. As an example, a web-based system for operating a virtual wind tunnel is presented. The authors offer some interesting solutions regarding effectiveness of voice command recognition e.g. list of synonyms that provide a more natural way of voice communication and so-called "execution words" that let recognize the end of voice command. However, their system, based on the limited details provided, appears to be not a remote control system in a strict sense. Although it is web-based, all voice communication seems to occur at the client side because all system components are packaged as activeX controls that can run on a web page. There is no communication with any real remote devices.

Ayres and Nolan [48] present voice activated command and control system for the real remote devices. Prototype application is used for controlling Lego robot. The authors consider various architectures of remote voice control system, each based on different distribution of system components at server side and client side. However, their approach is based on a small set of simple voice commands. Besides, their solution is not industrially oriented. Therefore they did not have to perform a detailed analysis of how the system's architecture would be influenced by processing voice commands in quasi-natural language. They also did not consider specific features of industrially oriented

voice control systems. Such analysis is the subject of the current paper.

4. General remote voice control scenario

According to what has been written in Section 2, a very promising area of industrial implementations of voice control systems are manufacturing cells of diversified automation level, where some functions are fully automated but there are also actions that must be taken manually by the operator. This paradigm cannot be applied to the remote voice control because no direct man-machine cooperation takes place. Nevertheless some factors (listed in Section 2) determining potential areas of industrial voice control implementations may also refer to the remote control scenario. Let us analyze them consecutively.

One of the important advantages of voice communication is that it does not engage operator's hands. Although the remote operator is not directly involved in any manufacturing activities, his hands may be engaged in entering data. In order to increase effectiveness of man-machine remote interaction, parallel voice communication can be applied. Following examples illustrate this idea:

- As stated before, the visual feedback plays an important role in industrially oriented remote control systems. The operator may control remote robots and other machines traditionally i.e. using keyboard and mouse, but the voice commands could be used for parallel control of visual feedback e.g. changing zoom, switching between cameras etc.
- Manufacturing task is often described by big amount of data that requires many input fields on the website form to be filled in. The user may point with the mouse cursor to the individual input fields and then use utterances to fill those fields with data. It may increase the effectiveness of data input, particularly when numerical values are involved e.g. dimensions of the workpieces.

Video 1 presents industrially oriented application of combined mouse–voice data input in the remote operation.

Another advantage of voice communication is that it does not restrict operator's mobility. As opposed to the direct control of

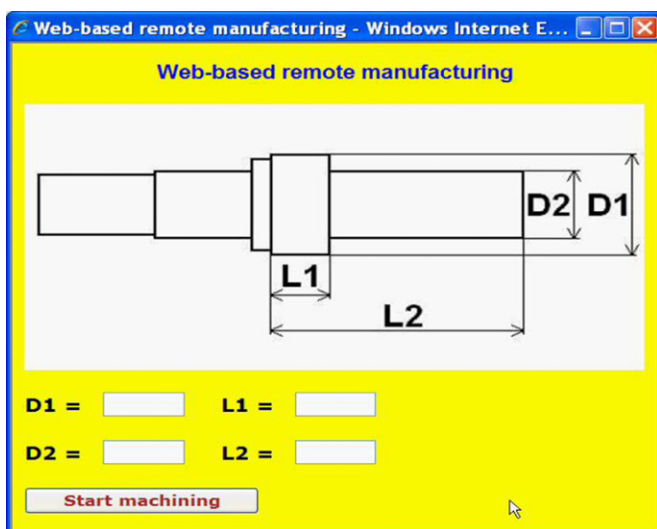
industrial equipment, the remote control does not require the user to move (in order to supervise several machines and/or devices). Visual feedback from individual machines can be obtained on computer screen by switching between cameras. In this way the operator's "virtual mobility" is granted. However, the real mobility is restricted because the user has to sit before computer screen (to obtain the visual feedback) and operate the keyboard and/or the mouse (to enter data). We can imagine a scenario where operator's mobility within his/her room is not restricted by the visual feedback because a big screen showing remote manufacturing process is installed there. In order to overcome mobility restrictions imposed by the way of data input (keyboard, mouse), the voice communication through wireless microphone can be applied (see Fig. 1).

As stated in Section 2, uttering a voice command in quasi-natural language may be more effective (simple and quick) than traditional input of complex data. This is true also for data input using forms on the web pages. The use of voice commands would be particularly advantageous if information derived from image recognition system were used to clarify ambiguities in the voice command (hence the voice commands could be then articulated in a compact way).

Let us imagine there are several workpieces of various types on the pallets and there are several machine tools used for machining those workpieces. There is also a robot responsible for loading and unloading those workpieces. The use of voice commands like: "Robot, load the shaft from the third pallet to the milling machine" could replace a web page containing a sizeable set of input elements (e.g. radio buttons).

The remote voice control scenario is also determined by following remarks:

- As stated in Section 3, it is preferable that the remote control be based on task-oriented options corresponding to complex actions taken by machines. Such complex actions can be



Video 1. Industrially oriented application of combined mouse–voice data input in the remote operation. A video clip is available online. Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.rcim.2012.11.002>.



Fig. 1. Operator controlling the remote robotized cell with voice commands. For better picture quality, adjustments of brightness and contrast have been applied to the area showing the screen.

described mainly by voice commands in quasi-natural language. Therefore the speech recognition system should be able to process the commands of that kind.

- According to Lan [49], web-based systems let the small and medium size companies use remotely operated machines without owning them. Therefore we may assume that the remote operators will usually use only a part of system resources (according to their current needs). This remark is crucial for voice-based remote control systems: as the speech recognition reliability is worse for sublanguages using very big vocabularies, the set of possible voice commands should be restricted to those really needed in particular case. One of solutions to that problem is to let the user define the voice command language. The remote voice control system architecture should make it possible in simple and user-friendly way.
- Additional advantage of solution proposed in the previous point would be the possibility to adjust the language to user's customs as well as to use various national languages in the same system.

5. Voice control system architecture

In order to realize scenario presented in Section 4, it is necessary to develop the architecture of industrially oriented voice control system that could be combined with web-based remote control. The first step is to specify the voice command processing scheme. Based on this, it will be possible to decide how to distribute individual components between server and client side in order to provide effectiveness and user-friendliness.

As stated in Section 4, the remote voice control scenario should be based mostly on voice commands in quasi-natural language. As proposed in Section 2, this can be obtained using VCD format for voice command language description. VCD is characterized among other things by the following features that may influence system architecture [15]:

- Common description of syntax and semantics has a textual form. Therefore it will be relatively easy for the user to define the voice command sublanguage (according to what has been proposed in Section 4).
- VCD-based language description created by the user can be automatically converted into grammar rules used by SR engines developed by others e.g. into XML-based grammar rules used in Microsoft SAPI.

Voice command processing consists of the following main steps (see Fig. 2):

- Recognition of individual words (tokens) of the voice command by SR engine using grammar rules.
- Syntactic analysis using spreading activation algorithm [50] operating on the sequence of those tokens in order to extract the syntax tree from grammar graph. Under the term “grammar graph” we will understand a graph representing syntactic part of VCD-based language description (see explanation later in this section).
- Semantic analysis based on procedural semantic network. Its aim is to determine the identifier (e.g. name) and parameters of so-called execution procedure. Execution procedure contains

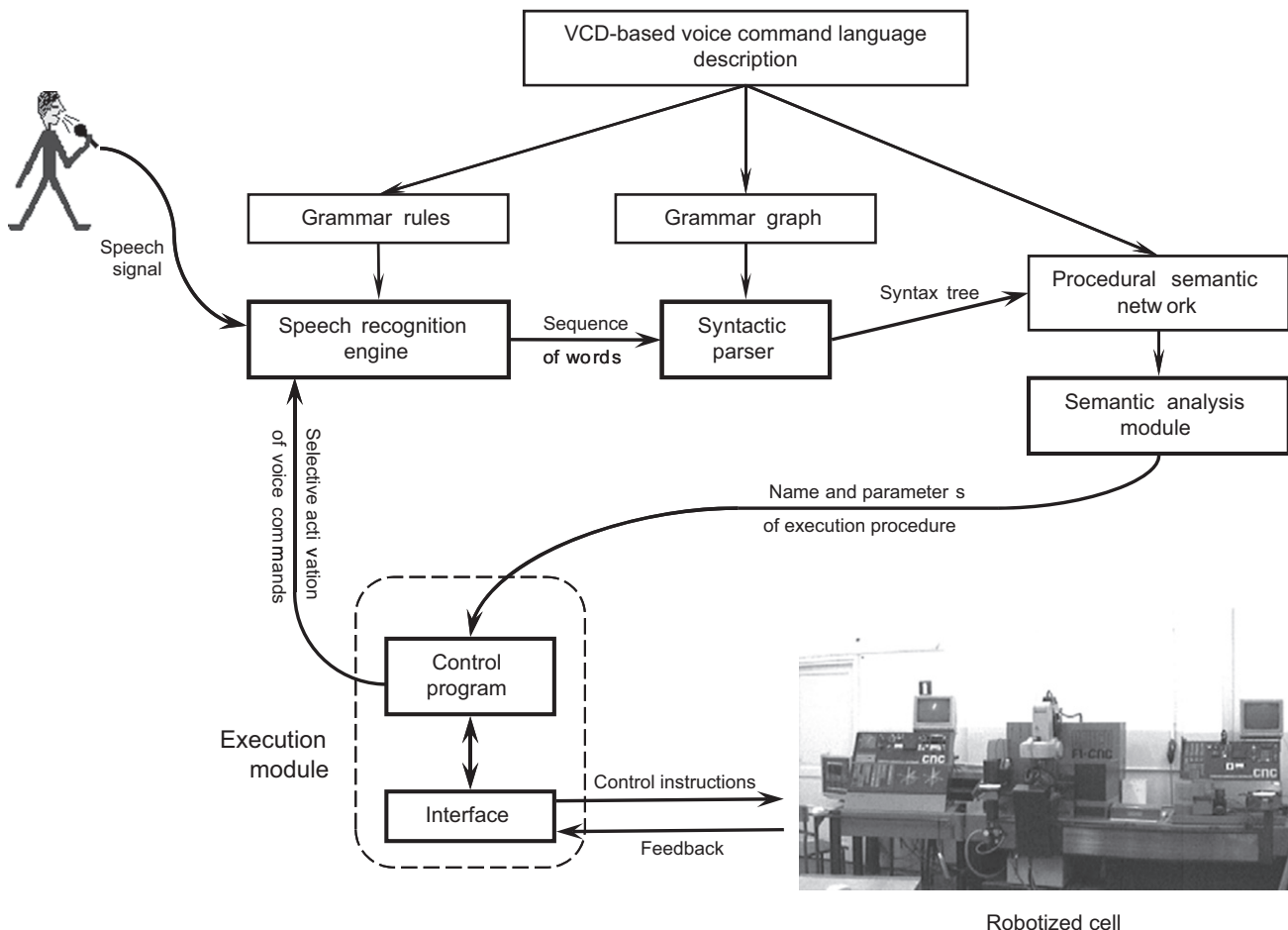


Fig. 2. General scheme of data processing in the remote voice control system.

a sequence of control instructions that are responsible for performing the task by machine.

As shown in Fig. 2, the VCD-based language description can be used as a basis for automatic construction of the following data structures required at individual stages of voice command processing:

- Grammar rules required by SR engine. As an example, a very simple VCD-based definition of sublanguage of voice commands used for robot control is presented below (for clarity reasons and in order to save the space, this example contains only very simple description of the phrase representing numerical values). These commands result in lifting various objects to the specified height:

```
#com lift
lift *object to the height of *dimension: p2, p7
#def object
the rod: 1
the shaft: 2
#def dimension
*value *units: p1*p2
#def value
ten: 10
twenty: 20
thirty: 30
#def units
meters: 1
decimeters: 0.1
centimeters: 0.01
millimeters: 0.001
```

The corresponding grammar rules in XML (used by Microsoft SAPI SR engine) can be automatically generated in the following form:

```
<grammar langid="409">
<rule name="lift" toplevel="active">
<phrase>
lift <ruleref name="object"/> to the height of <ruleref
name="dimension"/>
</phrase>
</rule>
<rule name="object">
<list>
<phrase> the rod </phrase>
<phrase> the shaft </phrase>
</list>
</rule>
<rule name="dimension">
<phrase> <ruleref name="value"/> <ruleref name=
="units"/> </phrase>
</rule>
<rule name="value">
<list>
<phrase> ten </phrase>
<phrase> twenty </phrase>
<phrase> thirty </phrase>
</list>
</rule>
<rule name="units">
<list>
<phrase> meters </phrase>
<phrase> decimeters </phrase>
<phrase> centimeters </phrase>
<phrase> millimeters </phrase>
</list>
</rule>
</grammar>
```

- Grammar graph used by syntactic parser. The grammar graph represents all possible variants of all parse trees that can be obtained as result of syntactic analysis of voice commands. The non-terminal nodes of the grammar graph represent phrases constituting voice commands. The terminal nodes represent individual words or elementary phrases (an elementary phrase is a phrase expressing elementary information hence it does not have to be divided into individual words e.g. "the workpiece"). Two kinds of edges are used: the solid lines connect the phrases with their components (lower-order phrases and/or individual words), whereas the dashed lines connect the phrases with their possible variants. Fig. 3 presents the grammar graph corresponding to the definition of sublanguage presented in the previous point (the names assigned to the phrases are written in uppercase, whereas the individual words are written in lowercase). As a result of syntactic parsing, the syntax tree can be extracted from the grammar graph. The part of grammar graph constituting syntax tree corresponding to the token sequence "Lift the rod to the height of twenty centimeters" is emphasized in Fig. 3 using thick lines and bold text.
- Procedural semantic network used in semantic analysis. This network is automatically constructed combining the information derived from the parse tree and from the semantic data contained in VCD-based sublanguage description. The nodes of semantic network (built upon the nodes of the syntax tree) are assigned one or several numerical values ("node parameters") representing information contained in phrases corresponding to those nodes [15]. Terminal nodes are assigned constant values only. Non-terminal nodes are assigned also procedures used for calculation of their node parameters (those procedures are practically reduced to arithmetic expressions built using parameters of children nodes). Semantic network built upon the parse tree corresponding to the voice command "Lift the rod to the height of twenty centimeters" (extracted from grammar graph shown in Fig. 3) is presented in Fig. 4.

The voice control system architecture can be also influenced by necessity to introduce the mechanisms providing improvement of speech recognition reliability. Contemporary SR engines are still far from being perfect. Various techniques are used in order to

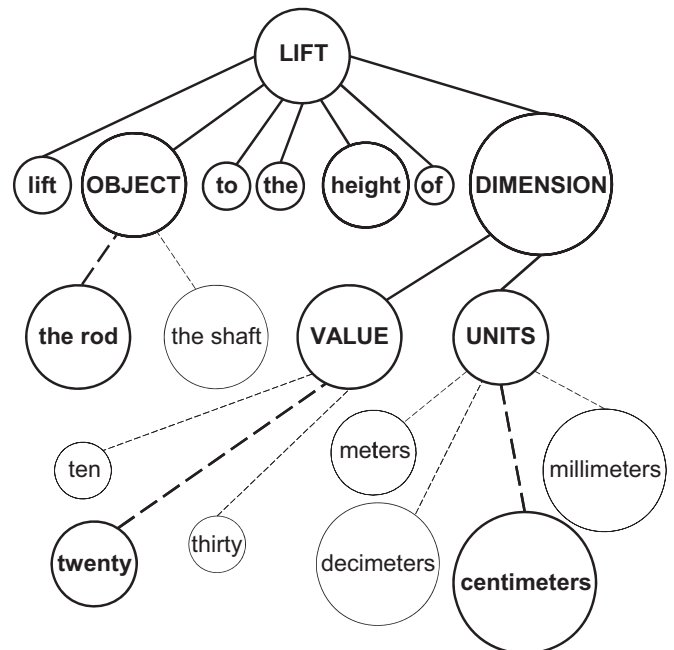


Fig. 3. Sample grammar graph representing syntax of voice command sublanguage.

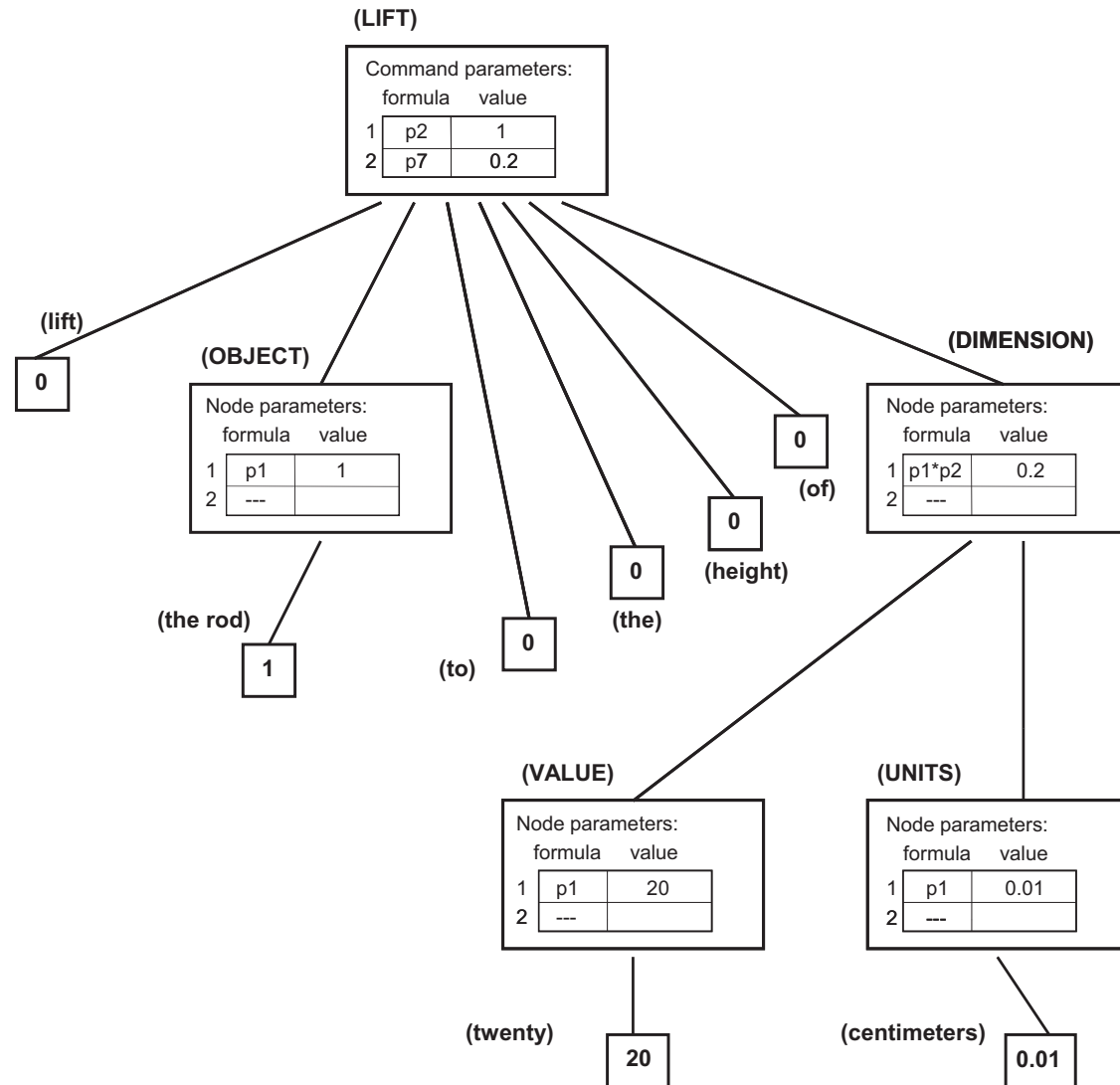


Fig. 4. Sample procedural semantic network used for semantic analysis.

compensate for this fact. One of them is appropriate use of past clarification dialogs [44]. Another one is analysis of past user behavior and prediction of the most probable commands under given circumstances [51]. Probably the most effective method that could be used in industrially oriented voice control systems is selective activation of voice commands. Depending on the current state of the robotized cell it is usually easy to determine which commands are appropriate in the current context. Therefore the program running within execution module may not only control the machines and devices but also send messages to the SR engine in order to determine which commands should be currently activated (see Fig. 2). In this way the number of utterances expected by the SR machine can be substantially reduced. Therefore a better speech recognition accuracy may be obtained.

6. Distributed voice control of robotized cells

In order to design a web-based voice control system it is necessary first to decide how to distribute its individual components (shown in Fig. 2) between server and client side. Solutions presented by researchers who deal with distributed speech recognition [36–39] often assume the use of mobile devices with limited processing power at the client side. Therefore the speech

recognition is usually performed at server side. Such scenario is also considered by Ayres and Nolan [48] as appropriate for a voice control system. However, according to conclusions presented in Section 3, industrially oriented remote voice control systems require also some form of visual feedback to be provided. On the other hand, the remote control system architecture should provide minimization of the network traffic. Therefore previously mentioned solution proposed by Wang et al. [24] can be applied. It consists in representing the remote machines and devices by a Java 3D model at the client side. This requires, however, big processing power. For big processing power should be available at the client side, it would be reasonable to install also the speech recognition engine there (in order to avoid sending speech signal over Internet). Now, the question must be asked: what about the other steps of voice command processing? Looking at Fig. 2 it seems reasonable to leave only the execution module at server side. The network traffic would be then reduced to sending calls of execution procedures and receiving signals related to selective activation of voice commands. However, there is also a disadvantage of such system architecture: the necessity to install all the software responsible for voice command processing at the client side. Let us notice that this software is partially not general-purpose, but application specific. Any potential modifications/improvements within the robotized cell control system would

require the software at client side to be updated. If the syntactic and semantic analysis were performed at the server side, such changes could be easily introduced without necessity to involve the remote users. In order to operate the system, the users would need only to install the universal software: a web browser and SR engine. This would significantly increase the portability.

Let us analyze the latter variant in detail. The network traffic would be generally restricted to sending recognized sequence words and receiving signals related to activation and deactivation of voice commands. However, the following remark seems to be a source of some inconvenience: the grammar rules, needed by SR engine are needed at the client side, whereas the grammar graph and procedural semantic network are needed at the server side. All these data structures are generated on the basis of the same sublanguage description in VCD. It means that exactly the same VCD file must be present at both sides. However, this should not constitute a problem because, as proposed in Section 4, it is the user who defines the voice command sublanguage and describes it in a textual form. The file containing VCD-based sublanguage description can be sent to the server prior to the system start-up.

The control program running within the execution module can be created either by the system designer or by the user. Both solutions have their advantages. The first version does not require the user to deal with programming at all. The control program offers a set of pre-defined task-oriented options that can be chosen by the remote user by voice. The second version offers more flexibility because the user may customize the system according to his/her needs.

Summarizing, the remote voice control system start-up is preceded by the following preparation steps (Fig. 5):

- The user defines the voice command sublanguage and describes it in VCD format.
- The control program is created by the user in a textual form (optionally).
- Using the system web page, those files are uploaded to the server.
- Based on the uploaded VCD-based sublanguage description, the grammar rules for SR engine are automatically generated at server side.
- The file containing grammar rules is downloaded to the client's computer.
- The control program starts.

The system operation itself comprises the following steps that can be repeated many times (Fig. 6):

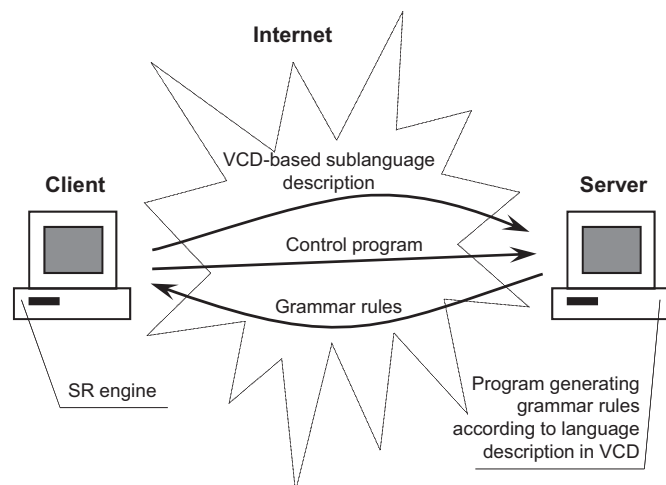


Fig. 5. Preparation to the remote operation.

- The control program sends a web page (html file) that contains voice-activated components. Those components (e.g. text input fields) are associated with the file containing grammar rules (previously downloaded by the user). Information on selective activation of voice commands is also contained in the web page (see Section 7 for an example).
- Voice command is uttered by the user. The sequence of words (tokens) is recognized by SR engine and automatically entered into appropriate input field on the form contained in the web page. The form is automatically submitted.
- Syntactic and semantic analysis is performed at the server side. Depending on their results, either the proper instructions are sent to the machine's controller or the information about error (wrong command) is sent to the user.
- Visual feedback is provided either independently or as a part of the next web page sent to the user.

In order to provide effective voice communication, it is necessary to include various mechanisms for initial checking of the recognized sequence of words already at the client side. For example, when the operator has spoken something mistakenly, it would be good to be able to break (cancel) the currently spoken command in order to speak it anew. For this purpose, so-called escape sequence (a sequence of words that are responsible for interruption of voice command recording) is used in VCD format [15]. Script responsible for handling hypothesis event in the SR engine should appropriately react to the presence of escape sequence in a partially recognized utterance. A sample thereof will be presented in Section 7.

7. Laboratory implementation

The remote voice control system architecture presented in the current paper was implemented on the educational robotized manufacturing cell in the Laboratory for Automated Manufacturing at the Warsaw University of Technology (<http://www.cim.pw.edu.pl/arog>). This cell consists of two machine tools (lathe and milling machine), automatically operated fixtures, industrial robot (for manipulating the work-pieces) and pneumatic slide (for moving the robot from one machine tool to the other one). All machines and devices are controlled by the cell's main computer via RS-232. Technical measures that had to be taken in order to adapt the manufacturing cell for remote control have been presented in another paper [52].

Several cameras providing the visual feedback are placed within and around the cell. A part of the speech recognition system (responsible for syntactic and semantic analysis) runs on a separate computer and communicates with the cell's main computer over the network. The other part (SR engine) runs on computers of the remote users. As the tests carried out by Ayres and Nolan [48] proved that Microsoft SAPI was among the best products, characterized by very low recognition time and very high recognition accuracy, this software was used also in the implementation presented in the current paper. It is worth mentioning that many other researchers used it in their voice control systems [7,12,13,45]. Another argument for using SAPI is its integration into Windows operating system as well as the possibility to use it by scripting languages such as JScript. Therefore it is easy to create voice-activated web pages. The following JScript code is responsible for creating basic SAPI objects within the web page:

```

VoiceObj=new ActiveXObject("Sapi.SpVoice");
SpRecoContext=new ActiveXObject("Sapi.SpInProcRecoContext");
ObjTokCat=new ActiveXObject("Sapi.SpObjectTokenCategory");

```

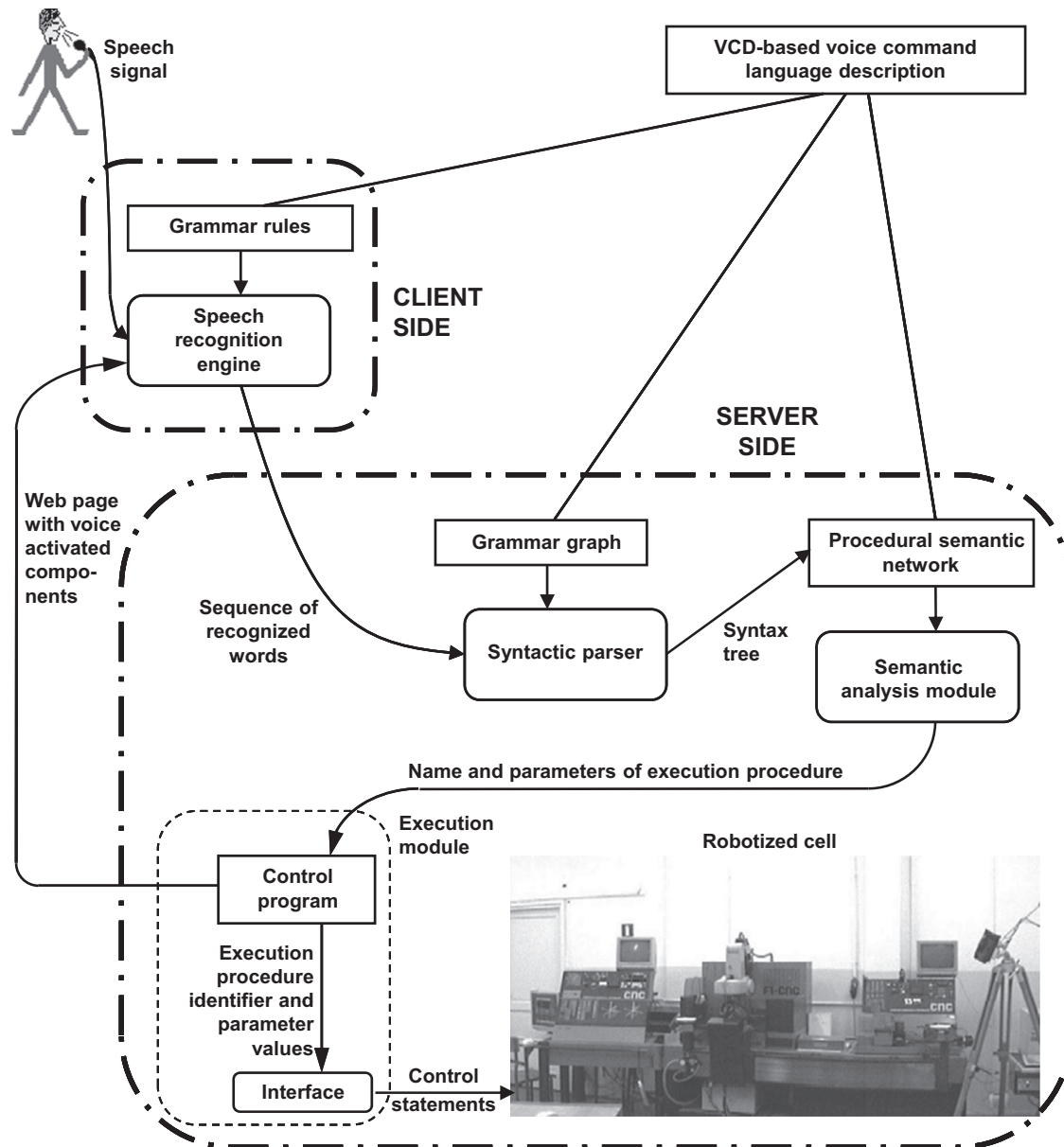



Fig. 6. Architecture of the web-based voice control system.

```
ObjToken=new ActiveXObject("Sapi.SpObjectToken");
SpLexicon=new ActiveXObject("Sapi.SpLexicon");
```

In order to prepare SAPI for operation, the following instructions are performed:

```
var recognizer=SpRecoContext.Recognizer;
var grammar=SpRecoContext.CreateGrammar(0);
VoiceObj.Priority=2;
ObjTokCat.SetId("HKEY_LOCAL_MACHINE\SOFTWARE\
Microsoft\Speech\AudioInput",0);
ObjToken.SetId(ObjTokCat.Default,"",0);
recognizer.AudioInput=ObjToken;
```

The following sample instruction is responsible for loading grammar rules contained in the file *commands.cfg*:

```
grammar.CmdLoadFromFile(filepath+"commands.cfg",1);
```

Individual web page components can be activated by voice in various ways. The following sample function submits the form after inserting the recognized voice command (string *recognized*) into the text input field *command*:

```
function recognize (recognized)
{form.command.value=recognized;
form.submit}
```

For user's convenience, partially recognized voice commands (hypotheses) can be also displayed on the web page. The following sample function realizes that scenario. It is also responsible for handling so called escape sequence i.e. the utterance resulting in the interruption of voice command recording (see Section 6). When the remote user says "Sorry, interrupt" (this is the escape

sequence used in this implementation) then the field *command* is cleared. SR engine state is set to inactive and all active audio data is purged. Subsequently, SR engine is activated again in order to record a new command:

```
function hypothesis(recognized)
{form.command.value = recognized;
if (recognized.indexOf(EscapeSequence) >= 0)
{form.command.value = "";
recognizer.State=3;
recognizer.State=1}}
```

The grammar rules contained in the above mentioned *commands.cfg* file are automatically created on the basis of VCD-based sublanguage definition. An excerpt from a sample *commands.cfg* file is presented below:

```
#esc Sorry, interrupt
#com camera1
camera one
#com camera2
camera two
#com camera3
camera three
#com camera4
camera four
#com unload
unload milling machine
#com place
place the workpiece in position *position of the *left_right pallet
#def position
three: 3
four: 4
#def left_right
left: 1
right: 2
```

As mentioned before, a selective activation of voice commands provides better recognition rate. Therefore the web pages used for remote control contain Javascript code responsible for that. For example, the following set of instructions is responsible for activation of all voice commands defined above except the one resulting in milling machine unloading:

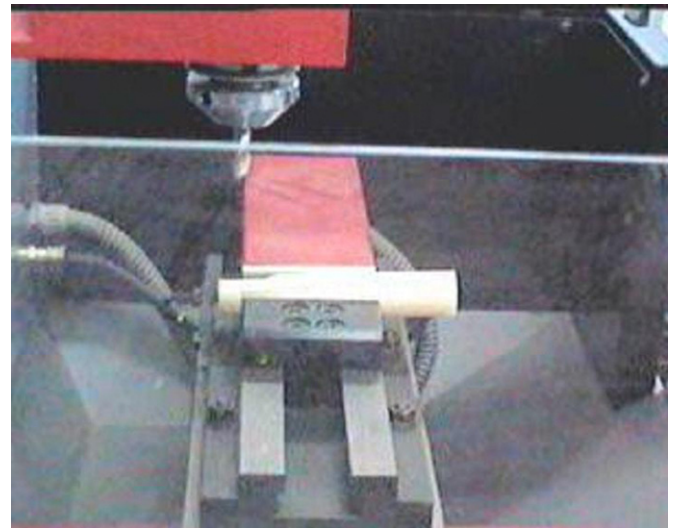
```
grammar.CmdSetRuleState("camera1",1);
grammar.CmdSetRuleState("camera2",1);
grammar.CmdSetRuleState("camera3",1);
grammar.CmdSetRuleState("camera4",1);
grammar.CmdSetRuleState("place",1);
```

All aspects discussed in the current section are illustrated by **Video 2**. It shows the remote operation of the robotized cell using voice commands in quasi-natural language.

Readers interested in more examples of complex voice commands in quasi-natural language are encouraged to read another author's paper published in "Robotics and CIM" [15] and to watch a video clip attached to the electronic version of that paper.

In order to provide full examples of quasi-natural language description in VCD, there is a supplementary data file attached to the current paper. This file contains following examples:

- Sublanguage of commands used for controlling a robotized cell at the Laboratory for Automated Manufacturing.
- Library of phrases representing numbers and arithmetical expressions. In order to demonstrate application of such library, a sublanguage of "voice calculator" is also presented.



Video 2. Controlling the robotized cell with voice commands in quasi-natural language. A video clip is available online. Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.rcim.2012.11.002>.

8. Experimental results

Practical verification of the method presented in the current paper comprises two main points:

- Speech recognition accuracy.
- Comparison between effectiveness of traditional (keyboard/mouse) and voice-based data input.

As far as speech recognition rate is concerned, substantially there is no difference between remote operation and direct voice control. Therefore there was no reason to conduct separate tests. The results obtained from the tests related to the direct voice control [15] are valid also for remote control. There is only one exception: the experiments involving noise produced by manufacturing equipment. As the remote users are practically isolated from this noise, it cannot influence the speech recognition rate.

Test results presented in the referenced paper show that the main source of limited voice control system reliability is deficiency of SR engines. Its influence can be, however, neutralized by implementation of appropriate mechanisms. Perhaps the most important one consists in providing the user with ability to check the *s* of the recognized voice command (hypothesis) before sending it to the server and to block the execution of falsely recognized commands. This mechanism was discussed in detail in the example presented in **Section 7**. So called escape sequence makes it possible to interrupt recording of currently spoken command and to record it anew. Although this mechanism usually effectively prevents execution of falsely recognized commands, it does not suffice to make the remote voice control system really effective and user-friendly. The necessity to repeat a command (if it is not an exceptional case) would make advantages of voice control system questionable.

As far as effectiveness of various methods of data input is concerned, the time needed to enter data was used as the yardstick of comparison. The following configuration was the basis for experiments:

- SAPI 5.1 speech recognition engine (pronunciation sensitivity 50%, accuracy vs. recognition time factor 50%).
- Windows Internet Explorer version 8 on client computer (Celeron CPU 3.07 GHz, Windows XP, Service Pack 3).

- Web page form containing 20 input fields for numerical data (mainly dimensions of the workpieces to be manufactured).

In order to compare the effectiveness, the user entered the same data (dimensions previously written on a sheet of paper) using three various methods:

- Mouse and keyboard.
- Mouse and voice (pointing to the appropriate field with mouse and uttering the phrase corresponding to the value to be entered).
- Voice only (specifying by voice both the field and value to be entered into that field e.g. “Dimension D is twenty nine point five”).

The profile of users who tested the system was following:

- System was tested in English by the users who speak this language fluently. However, there was not a native English speaker among them.
- Users' typing skills were good or very good (they use computers in everyday professional practice).

The experiment was conducted ten times. The results are presented in Table 1.

The results are, of course, strongly influenced by individual user's abilities and skills. Occurrence of typing errors may significantly change the time needed to fill in the form on the web page. Nevertheless the advantage of voice input over mouse/keyboard input is apparent in the case of forms containing many input fields to be filled with numerical data. Besides, all users who took part in the experiment agreed that the voice input was more comfortable. The main disadvantage of traditional method (mouse/keyboard input) is the necessity to type in the numerical data and to frequently change the focus between keyboard, screen and the sheet of paper. The combined mouse/voice data input method is much more comfortable because the user needs to change the focus only between the sheet of paper and the screen. The third method (data input by voice alone) could be considered the most comfortable one because the user can (theoretically) look at the sheet of paper only. Practical experiments, however, do not confirm that. Because false recognitions occur time to time, it is necessary to trace the changes on the screen in order to check if they correspond to user's intentions. In the experiment described above, there were situations when the numerical value corresponding to user's utterance was inserted into a wrong field (because the names of two fields were similar). Sometimes it was even difficult to find which field has been updated in that way. Therefore we can draw a conclusion that the

combined mouse/voice input method is the most suitable one in the case of forms containing many numerical input fields.

A separate topic is the reliability of remote communication based purely on complex voice commands in quasi-natural language. As stated by Rogowski [15], the reliability of voice control systems is limited mainly by deficiency of speech recognition engines and by the presence of noise. In the case of remote control, the users are practically isolated from the noise produced by manufacturing equipment. Nevertheless false recognition is highly possible e.g. when similarly sounding words like “fifteen” and “fifty” may be expected in the voice command. In such a case, the system should inform the user what it has understood (before the action is taken by machine). In this way the user is provided with the possibility to break the command and speak it anew. This mechanism can be easily implemented in remote voice control system because the contents of partially recognized voice command (hypothesis) may be instantly displayed on the web page, hence the user may easily check whether the recognition was correct (and react appropriately). The example thereof was presented in Section 7.

Few words should be said about time delays related to the remote voice control. Following factors may influence the time elapsed between voice command and its execution in machine:

- voice command recognition and analysis,
- data transfer between client and server,
- generating NC code,
- loading NC program to the controller.

The time needed for voice command recognition and analysis is so short that it can be practically ignored (in the worst case it did not exceed 0.25 s). As far as NC programs are concerned, there are two possible scenarios: either the voice command results in running one of previously prepared NC programs or that program must be first automatically generated according to the contents of the command. In the first case the machine reaction was almost immediate. In the second case there were delays up to 1.5 s. Those delays were measured for direct (not remote) voice control of the robot performing short sequences of manipulation tasks (e.g. workpiece transfer from the pallet to the machine tool).

Due to the proper architecture of remote voice control system presented in this paper, the amount of data flowing between client and server is minimal. Therefore time delays caused by data transmission did not introduce big impediments in remote voice control. Overall time elapsed between voice command and its execution in machine never exceeded 3 s. However, it must be emphasized that this parameter may generally vary in a broad range, depending e.g. on the current Internet connection speed.

In the case of interruption of previous voice command recording, some time is additionally needed to initialize a new recording. As it can be seen in video clip referenced in previous section, this additional time delay does not exceed 0.5 s.

Table 1
Overall time (s) needed to enter data into the web page form using various methods.

Data set	Mouse+keyboard	Mouse+voice	Voice
1	80	75	75
2	75	69	70
3	78	67	64
4	72	64	64
5	71	59	62
6	75	63	69
7	73	62	67
8	77	63	65
9	75	66	69
10	74	65	64
Average	75	65	67

9. Discussion

The main contribution of research presented in this paper to the area of remote control is the application of speech recognition technique in the web-based control of autonomous robotized cells. According to author's knowledge, it is the first such attempt. Only few works tried to cope with the problem of how to combine solutions in the field of speech recognition and remote control. None of them was aimed at industrial applications. Besides, the use of a small set of simple voice commands was presumed. Solution presented in this paper provides mechanisms for processing voice commands in quasi-natural language. Therefore a detailed

analysis was performed in order to develop the system architecture providing the most appropriate distribution of system components between server and client. According to author's knowledge, this is the first time when such analysis has been performed in relation to processing of quasi-natural voice commands. This solution, based on voice command language description in VCD format, provides simple customization, particularly with regard to the sublanguage and national language to be used. System portability is also provided because there is no need to install any software at client side except the web browser and SR engine.

It is important that theoretical considerations were practically verified with successful implementation described in Section 7. This implementation included mechanisms providing full reliability of the whole system despite limited recognition rate provided by SR engine: selective activation of voice commands and interruption of mistakenly spoken commands.

Experimental results presented in Section 8 show that the use of voice commands in web-based control of manufacturing equipment may increase both effectiveness of remote communication and user-friendliness. This indicates that industrial applications should be possible in the next future. Nevertheless it is probably too early to draw detailed conclusions. Although research presented in this paper proved that error free functioning can be provided, but the occurrence of false recognitions may significantly decrease attractiveness of voice communication because it induces the necessity to repeat the voice commands. Therefore research aiming at increasing the reliability of SR engines must be continued.

The future plans comprise the following:

- Development of universal mechanisms that would facilitate the use of context in voice command analysis (reference to previous commands, use of pronouns, cooperation with image recognition system with the purpose of meaning disambiguation).
- Development of universal VCD libraries of phrases that could be used in any industrially oriented remote voice control systems.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.rcim.2012.11.002>.

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